

OntoLLMJudge: A Framework for Neurosymbolic Evaluation of LLM Generations

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Abstract

Large language models (LLMs) are increasingly utilized to support a wide range of tasks, often open-ended in nature and requiring specialized domain knowledge. However, evaluating outputs, generated by LLMs on such tasks, presents a fundamental trade-off: reliable assessment requires expert judgment, yet manual expert annotation does not scale. Existing automated approaches, such as reference-based metrics and LLM-as-a-judge paradigms, fail to capture actual quality of LLM outputs or inherit the opacity and bias of the models they assess. To address this trade-off, we propose OntoLLMJudge – a neurosymbolic framework that utilizes ontology engineering as a principled control layer over neural evaluation. Rather than leaving expert criteria implicit in prompts or delegating them entirely to a learned model, OntoLLMJudge relies on explicit criteria formalized in an ontology. This makes the evaluation criteria machine-readable, transparent, and independently adaptable by domain experts without modification of the prediction engine. The ontology thus serves a dual role: it informs the neural predictors that score the generated content, and captures the expert-defined criteria for the final score interpretation. We instantiate and validate OntoLLMJudge on the automated evaluation of LLM-generated explanations of ontology modeling mistakes. Our findings indicate that ontology-grounded fine-tuning of scoring predictors consistently improves classification performance over pre-trained models, leading to an average improvement in Macro F1-score across dimensions of 0.49 over pre-trained models and 0.10 over classically fine-tuned models.

Keywords

Knowledge Engineering, LLM Evaluation, LLM-as-a-judge, Neurosymbolic AI

1. Introduction

Large Language Models (LLMs) have gained considerable attention due to their strong capabilities across a wide range of tasks and domains [1]. Thus, rigorous evaluation of LLM-generated outputs is essential to ensure their reliability in real-world applications, and strengthen awareness of their strengths and limitations, particularly in critical domains such as law, healthcare, and education [2, 3, 4, 5]. Nevertheless, the adoption of LLMs in knowledge-intensive tasks, dependent on expert understanding and careful reasoning, has made the evaluation of their outputs challenging [6]. LLM generated outputs on such tasks are often (1) open-ended, with a single input admitting many valid outputs, (2) assessable across multiple dimensions (e.g., factual correctness, relevance, conciseness) and (3) reliant on expert domain knowledge criteria for their assessment.

One research field that particularly exemplifies this evaluation challenge is Knowledge Engineering (KE). A concrete KE task, recently explored utilizing LLMs [7] and producing outputs exhibiting the characteristics outlined above (1-3), is the explanation of ontology defects in the context of ontology evaluation. To illustrate the task, consider a knowledge engineer’s intention to model the following requirement in an ontology: *Every CardGame includes at least one CardComponent*. A common modeling mistake would be formalizing this as:

Class: CardGame

EquivalentTo: Game and includes some CardComponent,

ELMKE: Evaluation of Language Models in Knowledge Engineering, 4rd Workshop co-located with ISWC 2026, Bari, Italy

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i.e., defining equivalence between the class *CardGame* and a *Game* that includes a *CardComponent*. However, the modeling choice of using **EquivalentTo** rather than **SubClassOf** incorrectly strengthens the constraint by introducing an additional unintended implication that *every Game, which includes a CardComponent, is necessarily a CardGame*.

Such mistakes must be explained in a way that accurately reflects expert knowledge and quality dimensions relevant for concrete application scenarios. For instance, in a collaborative LLM-expert ontology engineering framework, concise explanations and straightforward correction suggestions may be preferred. In contrast, in tutoring scenarios, where students validate their ontologies with LLM support, explanations should guide learners towards understanding the mistake by providing sufficient context, clarifying used ontology elements and describing unintended consequences, while potentially omitting direct corrections to encourage cognitive engagement and independent reasoning. Thus, the assessment of such explanations must be guided by expert-defined criteria.

Yet, expert evaluation does not scale for real-world application scenarios, introducing latency, high costs and enormous manual effort [8, 9, 10]. Moreover, automated evaluation approaches relying on reference-based metrics reduce to surface-level lexical or semantic similarity, which correlates poorly with human judgment [11]. Finally, newly emerged LLM-as-a-judge strategies inherit the problems of the very models they are meant to assess [6].

To enable reliable and scalable evaluation, this work addresses the following research question: *How can automated evaluation of LLM-generated outputs be made scalable while remaining aligned with expert assessment criteria?*

To mitigate the trade-off between evaluation quality and scalability, we propose OntoLLMJudge – a neurosymbolic framework that utilizes expert-defined evaluation criteria. Figure 1 provides an overview of the proposed solution. OntoLLMJudge combines neural predictors with explicit expert domain knowledge through dedicated fine-tuning. Instead of embedding expert knowledge in prompts or delegating evaluation entirely to learned models, OntoLLMJudge represents assessment criteria explicitly in an ontology. This formal representation makes the assessment criteria machine-readable, transparent, and easily adoptable, while remaining decoupled from the neural prediction component.

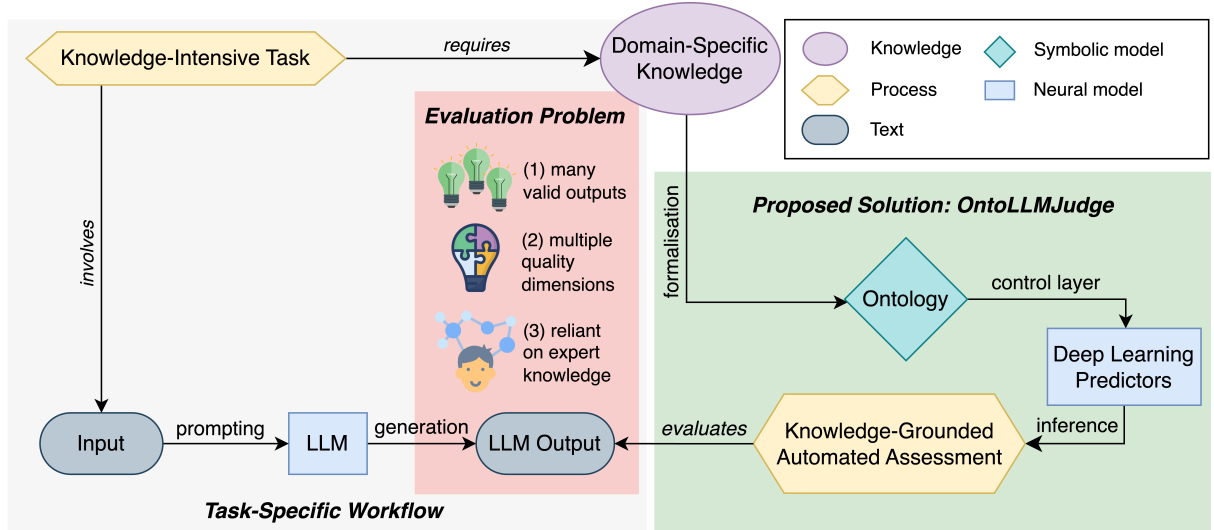


Figure 1: Overview of OntoLLMJudge. Left: the general evaluation problem for knowledge-intensive tasks. Right: our proposed solution OntoLLMJudge, where domain knowledge is formalized into an ontology that acts as a control layer over neural predictors during inference.

This work makes the following key contributions:

- C1.** We propose OntoLLMJudge, a neurosymbolic approach for the evaluation of LLM outputs, relying on an ontology for capturing expert evaluation criteria.

- C2. We instantiate OntoLLMJudge for a specific KE task, focusing on the assessment of LLM-generated explanations of ontology defects in junior-built ontologies. To this end, we create an ontology that captures multi-dimensional evaluation criteria previously utilized for the manual evaluation of this KE task in [7]. Finally, we utilize this ontology in the fine-tuning of a dedicated neural predictor per evaluated quality dimension.

Our evaluation results show that OntoLLMJudge outperforms pre-trained and classically fine-tuned models, yielding an average Macro F1 improvement of 0.49 and 0.10, respectively, across quality dimensions. We publicly share all resources, including source code, developed ontology, and used datasets online¹.

The remainder of this paper is structured as follows: Section 2 presents the OntoLLMJudge framework, including its ontology and neural components. Section 3 introduces the use case of LLM-based ontology mistake explanation, and Section 4 details its instantiation for this use case. Section 5 presents the evaluation, while related work is discussed in Section 6. Section 7 concludes the paper.

2. The OntoLLMJudge Framework

We propose a hybrid neurosymbolic approach whose central insight is that the domain-specific knowledge needed to evaluate LLM output is itself structured and can therefore be formalized. In KE settings, the criteria that determine whether a generated artifact is correct, complete, or pedagogically appropriate are not arbitrary: they derive from well-defined formal semantics, established modeling conventions, and task-specific expert judgment.

Framework Overview. Rather than leaving this knowledge implicit in prompt instructions or delegating it entirely to a learned model, we propose to represent it explicitly as an ontology, which then serves as a control layer over neural evaluation. A visual representation of our solution concept is presented in Figure 1. The left part of the figure illustrates the general problem setting: a knowledge-intensive task produces an input that is fed to an LLM, whose output must be evaluated. The three properties of the evaluation problem identified in the introduction recall why standard automated approaches are ill-suited in this context. The right part of the figure depicts the proposed solution, OntoLLMJudge. Here, the domain-specific knowledge is formalized into an ontology, which acts as a control layer over deep learning predictors. Together, they form a knowledge-grounded automated assessment process that evaluates the LLM output by combining symbolic reasoning and learned prediction.

Neurosymbolic Approach A key design principle of this framework is that expert involvement is structural rather than incidental. The ontology is not a static artifact generated as an auxiliary product of the system; rather, it is the primary component for representing the evaluation criteria, as well as any other relevant knowledge needed to support the definition of such criteria. This evaluation knowledge becomes actionable when we introduce a neural predictor—a pre-trained deep learning model—that connects LLM-generated content to its evaluation scores.

Indeed, with only the knowledge, we cannot automatically evaluate the content end-to-end; we still require a model for that. However, the neural predictors benefit from the structured guidance provided by the ontology, since their task—predicting the value of an evaluation criterion—is now informed by domain-specific expert knowledge, instead of relying solely on the general-purpose knowledge acquired during pre-training. The result is a system in which formal knowledge representation and learned prediction are complementary components of a knowledge-grounded automated evaluation framework.

¹<https://anonymous.4open.science/r/OntoLLMJudge/>

3. Use Case: LLMs as Ontology Evaluation Assistants

In this section, we introduce a concrete use case, for which scalable evaluation of LLM generated outputs is a significant challenge. In [7] the authors propose a number of ontology evaluation sub-tasks including the detection, classification and explanation of mistakes in ontology axioms. They perform an experimental investigation on the capabilities of LLMs to support these tasks in the context of developing an AI-assistant that can support students in building high-quality ontologies [12]. While the presented detection and classification tasks rely on the evaluation against a ground truth dataset, the evaluation of generated LLM explanation is performed manually by experts. To this end, an Expert-Assessment Schema (EAS) is defined, based on which each LLM explanation is evaluated.

The proposed EAS [7] captures four quality dimensions: (1) *Modeling comprehension* evaluates the interpretation of the modeled axioms and all relevant elements included; (2) *Inference capability* assesses the explanation of unintended outcomes as a consequence of the mistake; (3) *Mistake correction competence* evaluates the suggested mistake correction; (4) *Appropriate vocabulary usage* assesses the usage of appropriate terms. Each dimension is rated independently on a scale from -1 to 1, with scoring options varying across dimensions, as listed in Table 1. Finally, the dimension scores are summed to determine the overall quality. Additional details on each quality dimension and scoring level are available in [7].

Table 1

Scoring scale used to evaluate each quality dimension from the EAS [7].

Quality Dimension	Scoring Scale			
Modeling comprehension	1 (correct)	0.5 (partial)	0 (missing)	-1 (incorrect)
Inference capability	1 (correct)	0.5 (partial)	0 (missing)	-1 (incorrect)
Mistake correction competence	1 (correct)	0 (missing)		-1 (incorrect)
Appropriate vocabulary usage	1 (correct)		-1 (incorrect)	

To exemplify the usage of the EAS, we consider the example introduced earlier (Sect. 1) where equivalence is incorrectly used to express that “*Every CardGame includes at least one CardComponent*”, and an LLM generated explanation of this mistake:

Mistake Explanation: *By using owl:equivalentClass, the axiom incorrectly states that anything that includes at least one CardComponent is a CardGame, not just that every CardGame must include one. This can lead to unintended inferences where non-card games are classified as CardGame simply because they include a card component.*

Evaluation: The quality score is calculated as follows: *The model interpretation is correct (+1), the explanation of unintended consequences is correct (+1), a correction suggestion is absent (+0) and the used vocabulary is appropriate (+1).* Thus, the explanation quality score is 3.

To enable a scalable LLM evaluation on this task that preserves expert-knowledge grounding, we provide an implementation of the proposed neurosymbolic LLM content evaluation framework (see Sect. 2) for this use case and describe the concrete implementation in the following section.

4. OntoLLMJudge for Ontology Mistake Explanation

In this section, we present an instantiated solution of the proposed OntoLLMJudge framework (described in Section 2) for the use case of evaluating LLM-generated ontology mistake explanations (introduced in Section 3).

In Section 4.1, we describe the symbolic layer implementation, which consists of an ontology that formalizes the EAS, grounds inherently ambiguous evaluation concepts, and defines the output space of the predictors. Section 4.2 details the sub-symbolic layer, which consists of neural predictors that connect LLM-generated explanations to their evaluation scores, guided and initialized from the ontology.

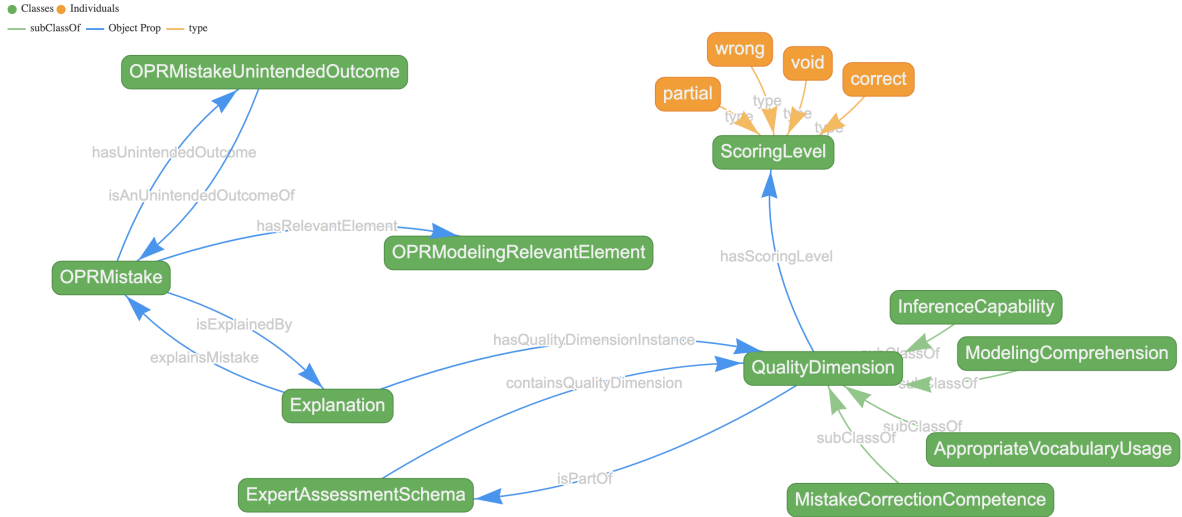


Figure 2: OrionBelt visualization of the ontology’s core classes (green) and individuals (orange). Green arrows denote subclass-of relations, orange arrows denote instance-of relations and blue arrows denote object properties relations. The mapping from ScoringLevel individuals to numeric score values is defined separately by the scoring strategy (see Fig. 3).

Throughout, OntoLLMJudge is designed to remain expert-in-the-loop, so that evaluation criteria and scoring logic can be freely adapted without altering the underlying modeling pattern. In Section 4.3, we provide an overview of the adaptability of the OntoLLMJudge implementation through additional expert knowledge.

4.1. Ontology Design

Core concepts and object properties. A visualization of the foundational concepts along with the defined object properties is given in Figure 2. The ontology revolves around three interconnected concepts: Explanation, QualityDimension, and OPRMistake. An Explanation is linked to the OPRMistake it addresses via the `explainsMistake` property, and to its assessed QualityDimension instances via the property `hasQualityDimensionInstance`.

For each explanation, four QualityDimension subclass instances are associated, each receiving a score that is an instance of the ScoringLevel class. As explained in Section 3, the EAS has four possible scoring levels: correct, partially correct, incorrect, or missing, formalized as the instances `correct`, `partial`, `wrong` and `void` respectively.

The four quality dimensions from the EAS are modeled as subclasses of QualityDimension, each further specialized into named subclasses encoding the possible assessment outcomes. A key design choice is that each such subclass is defined as an OWL intersection with a `hasScoringLevel` `hasValue` restriction, making the mapping between assessment outcomes and scoring levels explicit and machine-readable rather than implicit in the annotation. This allows a reasoner to classify a QualityDimension instance into the appropriate outcome subclass once its score is asserted. This step is performed when the neural predictors return their scores in the final stage of the pipeline.

The OPRMistake class is organized into seven mutually disjoint subclasses corresponding to pre-identified mistake types, available for the use case and described in detail in [7]. The disjointness axiom is a deliberate modeling choice: it enforces that each mistake belongs to exactly one category, which is a necessary assumption for the mistake-dependent formalization introduced below. OPRMistake is further connected to OPRModelingRelevantElement and OPRMistakeUnintendedOutcome, whose roles are described in detail below.

¹<https://orionbelt.streamlit.app/>

Ontology Extension. We extend the formalization by grounding two concepts which are needed for the application of the EAS but carry inherent ambiguity:

- (1) An understanding of *Relevant elements (REs)* is needed to determine whether a mistake explanation references all (or only a subset of) ontology elements included in an erroneous ontology axiom, that are connected to the encountered mistake (ModelingComprehension quality dimension);
- (2) *Unintended outcomes (UOs)* must be known to assess if an explanation correctly and completely points out unintended consequences of the encountered mistake (InferenceCapability quality dimension).

In both dimensions, the scoring outcome thus depends on whether the explanation accounts for all or part of these elements, which makes their explicit definition a prerequisite for a rigorous formalization. We leverage the mistake type classification to achieve this, as each OPRMistake subclass carries an expert-defined textual description (available from [7]) from which REs and UOs can be derived. To keep this step as grounded and expert-centered as possible, we develop two distinct processes for each concept.

First, relevant elements are derived automatically from the `rdfs:comment` of each OPRMistake subclass. Using the SpaCy library, each comment is tokenized and lemmatized, and the resulting lemmas are matched — using both exact and fuzzy matching — against a curated subset of the OWL specification taxonomy provided by the W3C². The matching results are then asserted as instances of `OPRModelingRelevantElement` and assigned to the corresponding mistake type via the `hasRelevantElement` object property.

Second, unintended outcomes are defined through an expert-validated list established for each mistake type. Rather than being asserted directly, the UOs of a given mistake instance are obtained by inference, meaning that the association between a mistake type and its unintended outcomes is encoded in the ontology and derived at runtime.

Ontology Evaluation. The developed ontology was evaluated through a combination of expert review by one of the authors of the original EAS and automated assessment using the OOPS! ontology pitfall scanner [13]. Any issues identified during the evaluation were addressed, and the final ontology is free of pitfalls according to the OOPS! catalog.

4.2. Neural Predictors

The sub-symbolic level of OntoLLMJudge is implemented as follows. For each of the four quality dimensions, we train a dedicated neural predictor — a generative deep-learning model — whose task is to determine which EAS scoring level best describes a given explanation.

Each predictor receives as input a structured prompt combining (1) an ontology axiom, (2) a modeling intention, (3) a detected mistake, and (4,5) the inferred REs and UOs. It produces a classification output corresponding to one of the scores defined for that quality dimension. For instance, predicting the score on *Modeling comprehension* is a four-class classification task with the possible outputs being one of correct, partially correct, incorrect, or missing.

This score prediction is then incorporated back into the ontology as a scoring instance, which is further associated to a score value depending on the ontology definitions.

Ontology-grounded output initialization. A central design choice is that the output space of each predictor is not defined arbitrarily but derived from the ontology itself. For each statement, a dedicated classification token is introduced into the predictor’s vocabulary. For example, *Modeling comprehension* has four possible outputs, so the vocabulary of the predictor is extended to four new tokens representing these four possible outputs. These new tokens come with their own embeddings, which are initialized with the embedding of the word describing the output.

²<https://www.w3.org/TR/owl2-syntax/>

For instance, the embedding of the token that represent the correct output is initialized with the embedding of the word "correct". This initialization provides a semantically grounded starting point that reflects the ordinal structure of the EAS scoring levels and ensures that the output space is aligned with the symbolic layer from the outset, before any task-specific training has occurred.

Subsequently, these embeddings are updated during fine-tuning so that they progressively move closer to the concept they represent—in this case, a correct comprehension of the modeling task. The newly introduced tokens also serve as the basis for the classification procedure. Indeed, at inference time, the predicted output corresponds to the token with the highest probability of being generated immediately after the prompt.

Fine-tuning technical details. Each predictor is fine-tuned using Low-Rank Adaptation [14], a parameter-efficient method [15] that introduces small trainable matrices into the attention layers of a frozen base language model, substantially reducing the computational cost of adaptation. Training uses a standard language modeling objective restricted to the single classification token position: the model learns to predict the correct statement token given the full evaluation prompt, with no gradient signal propagating from the prompt itself.

The training uses a weighted loss that assigns a higher penalty to errors on under-represented classes to account for possible imbalance present in the dataset. At inference time, the model’s output is constrained to the valid score level tokens for the task, so that each predictor always returns a well-formed EAS score that can be directly asserted into the ontology.

4.3. Expert-in-the-Loop Design Principles

A central strength of the OntoLLMJudge is the guidance by expert knowledge. Figure 3 presents a visual representation of the different stages of expert involvement in our system, represented as colorful blocks all around our system represented in the white block. While the use of neural networks is essential to our approach for deriving the final scores for the quality dimensions, grounding the ontology in expert knowledge is equally essential to assess the relevance of the whole automated evaluation process.

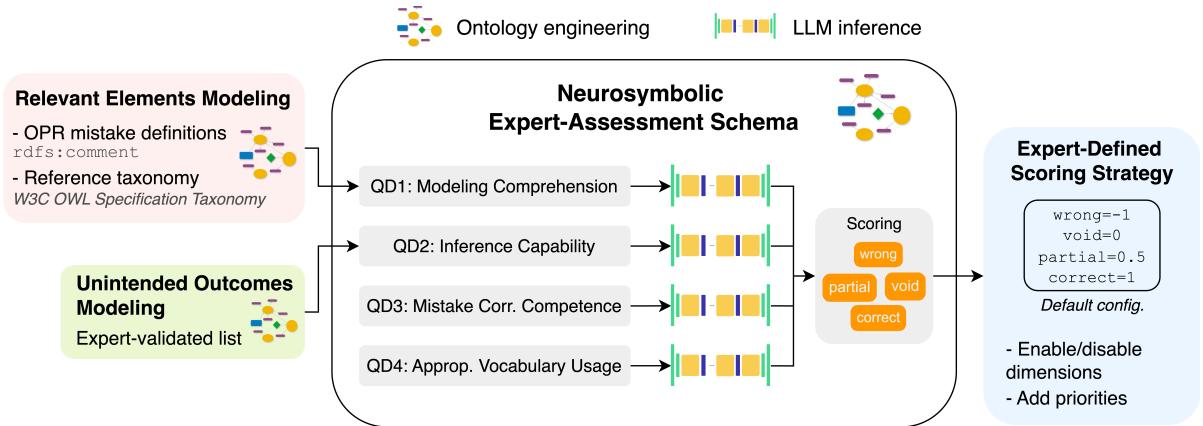


Figure 3: Expert involvement across the different stages of OntoLLMJudge workflow. QD1–QD4 denote the four quality dimensions, while icons mark whether a stage involves ontology engineering or LLM inference. The default scoring strategy comes from the Expert-Assessment Schema.

Expert evaluation criteria can be tailored to a new use case settings without modification of the sub-symbolic layer or overall ontology modeling patterns. The ontology concepts, descriptions, and scoring logic can be easily adopted or extended. For instance, additional mistake types can be introduced and scoring can be personalized such that score values are adjusted, quality dimensions are enabled or disabled, or different priorities between dimensions are introduced.

5. Predictor Classification Results

5.1. Experimental Setup

We train Qwen/Qwen3-30B-A3B-Instruct-2507 on the four tasks, an open-weight language model selected for its memory efficiency and competitive performance on generative tasks. Each predictor takes as input an enriched representation of the student input, combining the original axiom, intended meaning, and mistake type with the REs and UOs inferred at the second formalization level, formatted into a system prompt template suited to the model.

The dataset comprises real-world data provided by the authors of [7]. It consists of student-built ontology axioms, created as part of a beginner-level Ontology Engineering university course, each containing a modeling mistake. Mistake explanations were generated by both open-weight and proprietary LLMs of various sizes (Claude-3.7-Sonnet, GPT-4o, Deep-Seek-V1, Llama3.3-70B, Llama3.1-8B). For each explanation, an expert annotation following the EAS is provided. The dataset includes 200 samples, split into training (80%) and validation (20%) sets. All EAS quality dimension scores are represented except for one: the incorrect (-1) score for *Mistake correction competence* appears only once and was removed. The remaining class distribution is nevertheless significantly imbalanced across the quality dimensions, which we address through a weighted cross-entropy loss during training.

5.2. Results

We compare the proposed OntoLLMJudge (*Ontology-Guided FT*) approach against two baselines, as presented in Table 2: (i) a *pre-trained* model without further fine-tuning and (ii) a classical fine-tuned (*Classical FT*) model trained without ontological enrichment. We evaluate the four quality dimension predictors using class-wise macro F1-scores, reported alongside per-class support counts to contextualize performance under class imbalance.

The pre-trained baseline is evaluated using the exact same prompt as the fine-tuned predictors, without any parameter update. Its near-zero F1-scores on all but the dominant class reveal that the model systematically predicts *correct* regardless of the input – indicating that, without fine-tuning, the model has not sufficiently internalized the structure and semantics of the prompt to distinguish between quality statements, which confirms the necessity of the fine-tuning step.

Our fine-tuned predictors (Ontology-Guided FT) consistently outperform the baseline across all dimensions and classes. Performance is strongest on well-represented classes: *Modeling comprehension* and *Appropriate vocabulary usage* reach F1-scores of 0.92 and 0.97 on their majority class, respectively. Gains on minority classes are also notable – *Mistake correction competence* achieves 0.97 on the *missing* class (support: 38) – though rare classes can remain challenging, as illustrated by the *correct* class of the same dimension (support: 2, F1: 0.53). These results suggest that applying the same strategy on top of a stronger base model could yield further gains on underrepresented classes. Fine-tuning is also computationally lightweight, requiring approximately 5 minutes per epoch and 25 seconds for validation.

Impact of Ontology-Grounded Enrichment. To assess whether the improvements observed above stem specifically from the ontology-grounded enrichment rather than from fine-tuning alone, we compare our results to an intermediate baseline: *Classical FT*. In this case, the predictors are fine-tuned the same way than in *Ontology-Guided FT*, but their input prompt omits the REs and UOs, retaining only the original axiom, intended meaning and mistake type. This ablation isolates the contribution of the enrichment layer from the contribution of fine-tuning itself.

We restrict this comparison to *Modeling comprehension* and *Inference capability*, since these are the only quality dimensions whose scoring statements depend on the formalization extensions that are REs and UOs (see Section 4.1). For *Mistake correction competence* and *Appropriate vocabulary usage*, the EAS scoring criteria do not leverage enriched elements, therefore Classical FT and Ontology-Guided FT here boil down to the same experiment.

Table 2

Class-wise macro F1-scores on the validation set per quality dimension, for the pre-trained (PT), classically fine-tuned (Classical FT), and ontology-guided fine-tuned (Ontology-Guided FT) versions of Qwen/Qwen3-30B-A3B-Instruct-2507. Classical FT is reported only for *Modeling Comprehension* and *Inference Capability*, the two dimensions whose scoring criteria depend on ontology-derived relevant elements and unintended outcomes. Support gives the class distribution in the training and validation sets. Std. is computed over 5 cross-validation folds. **Bold** indicates the strictly higher score per row.

Quality Dimension	Class	Support		Macro F1-Score		
		Train	Val.	Pre-Trained	Classical FT	Ontology-Guided FT
Modeling Comprehension	correct	102	25	0.77	0.87 ± 0.03	0.84 ± 0.05
	partial	17	4	0.00	0.43 ± 0.16	0.58 ± 0.09
	missing	2	1	0.00	0.00 ± 0.00	0.00 ± 0.00
	incorrect	38	10	0.00	0.53 ± 0.09	0.62 ± 0.15
Inference Capability	correct	61	16	0.57	0.71 ± 0.04	0.69 ± 0.06
	partial	37	9	0.00	0.52 ± 0.07	0.53 ± 0.03
	missing	21	5	0.00	0.31 ± 0.09	0.61 ± 0.10
	incorrect	40	10	0.00	0.56 ± 0.11	0.60 ± 0.03
Mistake Correction Competence	correct	7	2	0.10	–	0.53 ± 0.14
	missing	152	38	0.00	–	0.97 ± 0.02
Appropriate Vocabulary Usage	correct	152	38	0.97	–	0.97 ± 0.01
	incorrect	7	1	0.00	–	0.00 ± 0.00

Ontology-Guided FT matches or outperforms Classical FT on almost every minority class across both dimensions. The only exceptions are the majority classes where Classical FT scores marginally higher, suggesting that the enrichment mainly benefits the less-represented classes. These results confirm that the observed gains stem specifically from the ontology-formalized REs and UOs, rather than fine-tuning alone, supporting our core hypothesis that explicit knowledge grounding improves predictor performance, especially on under-represented classes.

6. Related Works

LLMs are applied to a broad range of Knowledge Engineering (KE) tasks, making their evaluation a central concern, particularly when they interact with human learners in expert-knowledge domains such as KE education. This section covers three axes: LLM use in knowledge engineering, general approaches to evaluating LLM-generated content, and their limitations in expert-knowledge domains.

6.1. LLMs for Knowledge Engineering Tasks

Recent advances in LLMs have transformed KE tasks, particularly ontology engineering. In ontology learning from text, LLMs are used to extract types and properties to build initial conceptual structures [16]. In ontology completion, embedding-based and natural language inference approaches using LLMs are employed to generate missing axioms, while hybrid strategies show strong complementarity to prior graph-based methods [17]. Ontology alignment has similarly benefited from LLMs through zero-shot prompting, outperforming systems like BERTMap [18] on complex biomedical datasets when incorporating structural context [19].

More broadly, LLMs have been employed for a variety of generative KE tasks [20], including the generation of ontology axioms [21] and competency questions [22], ontology axiom evaluation [7], and ontology verbalization [23]. Despite these gains, LLMs are predominantly used as black-box generators, with ontologies serving merely as input or output targets rather than as active controllers of the process [24].

Yet, there is currently no standardized method to evaluate the quality of LLM-generated KE artifacts [25, 7], mainly because such outputs are open-ended, multi-dimensional, and rely on expert domain knowledge. Recent work on competency question generation [26], for instance, evaluates LLM-generated CQs across dimensions such as readability, relevance, and structural complexity, underscoring the multi-dimensional nature of KE artifact evaluation.

Some approaches structurally compare a generated artifact to a manually created resource, thereby assessing similarity rather than output quality itself [27, 28]. Another approach leverages the LLM-as-a-judge paradigm, relying on prompt engineering validated on only a small number of examples [29], which undermines the system’s reliability. We introduce an ontology-grounded control layer to address this gap, enabling structured evaluation grounded in formal expert knowledge.

6.2. General Approaches to Evaluating LLM Generations

Evaluating the quality of LLM-generated texts was initially performed using reference-based metrics such as BLEU [30] or ROUGE [31]. However, it has been shown that such metrics poorly capture output quality, as they only rely on semantic similarity and have been shown to correlate poorly with human judgment [11]. Learned metrics like BERTScore [32] and BARTScore [33] improve upon this, and more recently, the LLM-as-a-judge paradigm [8, 34] has been proven efficient with state-of-the-art proprietary models or after fine-tuning for domain-specific tasks [35].

However, overlap in training corpora amplifies bias, and this approach remains subject to the same black-box nature and hallucination risks as the models under evaluation. Several LLM-as-a-judge strategies have emerged to address these issues [34], but they introduce further complexity and lead to inconsistencies in findings and their interpretation [1]. Lastly, expert evaluation does not scale for real-world application scenarios, introducing latency, high costs, and demanding manual effort [8, 9, 10].

Human-in-the-loop approaches have emerged as a middle ground, enabling collaboration between humans and automated evaluators [36]. EvalGen [37] exemplifies this direction, but identifies a key limitation: some evaluation criteria are inherently output-dependent and therefore cannot be fully defined a priori, motivating approaches in which criteria are made explicit and formally grounded before evaluation begins. This is precisely the role of our ontology formalization.

6.3. Evaluating LLM Generations in Knowledge-Intensive Tasks

The limitations above are particularly acute in KE domains, where the correctness of LLM outputs cannot be assessed without grounding them in formal semantics. LLM judges are too sensitive to training data biases, leading to inconsistent judgments in knowledge tasks [38]. They also tend to exhibit hallucinated criteria [39], lack explainability [40], and fall short in interpreting domain-specific knowledge [41] — implying a critical inability to formalize ontology-specific constraints. In tutoring systems, existing approaches to evaluating LLM-generated feedback predominantly rely on this paradigm [42, 43], typically involving proprietary models prompted with expert-designed scoring rubrics [44]. PaCE [45] exemplifies such limitations: the assessment logic remains implicit in the prompt, evaluation criteria are neither formally defined nor reusable, and the evaluator model may share training data with the generation model, a situation known to introduce systematic bias [46]. Reference-based metrics are equally ill-suited, as the quality of an ontology mistake explanation cannot be reduced to lexical or semantic similarity to a reference text [47].

We address this gap by formalizing an expert assessment schema as an ontology, making evaluation criteria explicit and modifiable by domain experts independently of the prediction engine that scores the content.

7. Conclusion

Although Large Language Models (LLMs) are widely used across a variety of tasks and domains, evaluating their outputs remains a challenge. For open-ended knowledge-intensive tasks evaluation

is particularly difficult because multiple valid solutions may exist, assessment must consider multiple quality dimensions and reliable judgment depends on expert knowledge. Yet, expert annotation does not scale while automated methods fail to capture true output quality in such settings. To address the trade-off, we investigate the following research question: *How can automated evaluation of LLM-generated outputs be made scalable while remaining aligned with expert assessment criteria?*

Contribution and Findings. We present OntoLLMJudge - a neurosymbolic framework for the automated evaluation of LLM-generations. OntoLLMJudge leverages an ontology to capture expert assessment criteria, which is then used as a control layer over neural predictors. The ontology formalization makes output evaluation criteria explicit, machine-readable, and customizable, while allowing domain experts to revise scoring weights or extend the schema without altering the underlying modeling logic.

We instantiate the framework for a concrete knowledge engineering use case- the evaluation of LLM-generated explanations of ontology modeling mistakes. Our evaluation shows that OntoLLMJudge consistently improves over randomly initialized and pre-trained baselines, particularly for under-represented scoring classes. The presented results suggest that ontology-grounded initialization is a meaningful inductive bias for lightweight classifiers operating over structured expert knowledge.

Limitations. The instantiation of OntoLLMJudge relies on real use case data, which is of limited size. To preserve annotation quality, we deliberately avoid synthetic data augmentation, which consequently limits the training signal available to the neural predictors, particularly for under-represented scoring classes. As a result, despite meaningful improvements over pre-trained baselines, the predictors do not yet achieve the level of performance one would expect from a production-ready automated evaluator. However, we view the primary contribution of this work as the conceptual framework and its formalization, rather than the predictors themselves. The pipeline is fully open and designed to accommodate stronger models as they become available. Moreover, additional data are currently being collected to enable further experimentation and evaluation.

Future Work. The formalization of the EAS as an ontology not only enables automated evaluation of LLM generations but also provides a formal reward signal for generation control. The ontology’s scoring levels can support reward shaping in a reinforcement learning from human feedback or group relative policy optimization, where the generating LLM is fine-tuned to maximize scores on the EAS quality dimensions. This links generation directly to the evaluation criteria, enabling modular control over the four quality dimensions. In ongoing follow-up work we focus on this research direction.

Adaptability to other domains. Although, in this paper, we instantiate OntoLLMJudge for a single use case the framework can easily be adopted to other domains. In each new setting, it is sufficient to develop an ontology encoding the domain-specific evaluation dimensions and scoring criteria. While adapting the framework requires an initial effort from domain experts to formalize the evaluation criteria, this one-time investment enables scalable evaluation across future instances within the domain. The core framework architecture, where formal knowledge representation serves as the control layer over neural predictions, remains unchanged.

Acknowledgments

Declaration on Generative AI

During the preparation of this work, the authors used ChatGPT5 for: grammar and spelling check, paraphrase and reword, improve writing style. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the publication’s content.

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A. Online Resources

We publicly share all resources, including source code, developed ontology, and used datasets online at <https://anonymous.4open.science/r/OntoLLMJudge/>.